

Direct kinematics

Fundamentals of Robotics

University of Trento

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DISI-UNITN



Definitions on robotic manipulators

Direct Kinematics

The Denavit-Hartenberg Convention

Examples

- Three-link Planar Arm

- Spherical Arm

- Anthropomorphic Arm

- Spherical Wrist

Joint Space and Operational Space and Work Space

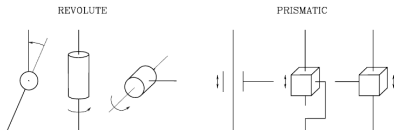
Useful definitions

Check quiz

Manipulators

Let us review some key definitions related to robotic manipulators

- ▶ A manipulator consists of a series of rigid bodies called *links*
- ▶ Links are connected by kinematic pairs called *joints*
- ▶ Joints are of two types
 - ▶ Revolute
 - ▶ Prismatic
- ▶ the two types of joints are graphically represented like this



Manipulators

- ▶ The whole structure forms a kinematic chain
- ▶ One end of the chain is connected to a base
- ▶ The other end is connected to an end-effector (gripper, tool) that is used for the manipulation tasks
- ▶ A kinematic chain is characterised by a number of degrees of freedom (DoF) that uniquely determine its *posture*
- ▶ The posture is defined by the poses of all the links
- ▶ Each DoF is associated with a joint variable.
- ▶ The goal of *direct kinematics* is to find the pose of the end effector given the joint variables

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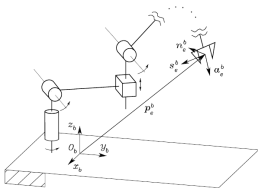
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Pose of the end-effector



$$T_e^b(\mathbf{q}) = \begin{bmatrix} R_e^b(\mathbf{q}) & \mathbf{p}_e^b(\mathbf{q}) \\ \mathbf{0}^T & 1 \end{bmatrix} =$$

$$= \begin{bmatrix} \mathbf{n}_e^b(\mathbf{q}) & \mathbf{s}_e^b(\mathbf{q}) & \mathbf{a}_e(\mathbf{q}) & \mathbf{p}_e^b(\mathbf{q}) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Pose of the end-effector

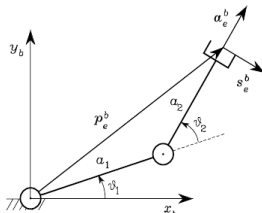
Given a fixed frame $O_b - x_b y_b z_b$ attached to the base and another attached to the end-effector, the pose of the latter is given by the homogeneous transformation matrix $T_e^b(\mathbf{q})$

- ▶ $\mathbf{q}$ is the n -vector containing the joint variables
- ▶ $\mathbf{n}_e, \mathbf{s}_e, \mathbf{a}_e$ are the unit vectors associated with the end effect and are chosen accounting for its geometry.
- ▶ If the end-effector is a two fingered gripper, $\mathbf{a}_e$ is chosen in the approach direction, $\mathbf{s}_e$ is chosen in the sliding direction and $\mathbf{n}_e$ is chosen orthogonal to the other two so that the frame $(\mathbf{n}_e, \mathbf{s}_e, \mathbf{a}_e)$ is right-handed.

All the elements of the matrix are functions of the joint variables $\mathbf{q}$.

Example

In simple cases, direct kinematics can be computed through straightforward geometric reasoning. Consider the example of a two link planar manipulator



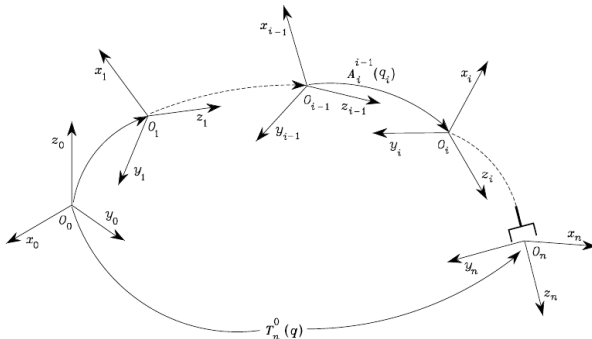
With basic calculations, we obtain:

$$T_e^b(\mathbf{q}) = \begin{bmatrix} \mathbf{n}_e^b & \mathbf{s}_e^b & \mathbf{a}_b^e & \mathbf{p}_e^b \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & s_{12} & c_{12} & a_1 c_1 + a_2 c_{12} \\ 0 & -c_{12} & s_{12} & a_1 s_1 + a_2 s_{12} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

($s_i = \sin(\theta_i)$, $c_i = \cos(\theta_i)$, $s_{12} = \sin(\theta_1 + \theta_2)$, $c_{12} = \cos(\theta_1 + \theta_2)$).

In more complex cases, direct derivation becomes challenging. Let's look

Open chains



- ▶ Let us consider an open chain with $n + 1$ links connected by n joints (no loops between the links).
- ▶ Link 0 is fixed to the ground
- ▶ By attaching a frame to each link the pose of the frame $i + 1$ will be determined from the pose of frame i through a simple transformation.
- ▶ The idea is that we can construct recursively the transformation between the 0-th and the n -th link recursively

$$T_n^0(\mathbf{q}) = A_1^0(q_1)A_2^1(q_2) \dots A_n^{n-1}(q_n)$$

where $A_i^{i+1}(q_i)$ is a homogeneous transformation matrix and is a function of the joint variable q_i

- ▶ To complete the picture, we need two additional transformations: one from the base to the first link and one from the last link to the end effector (which are typically constant)

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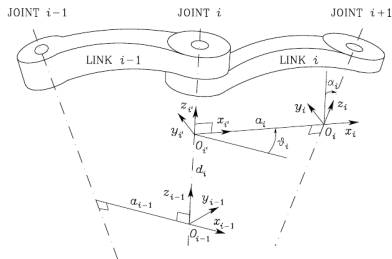
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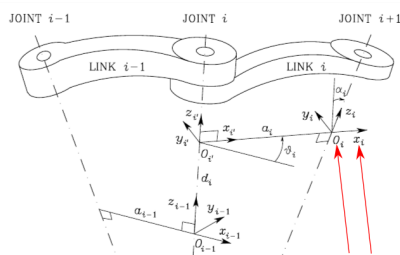
Denavit-Hartenberg Convention

- ▶ In order to find the direct kinematics of an open chain, we need an effective way to place the different frames
- ▶ The D-H method provides a solution
- ▶ Suppose that axis i connects link i with joint $i + 1$
- ▶ Axis z_i is chosen aligned to the Axis i
- ▶ We still have to choose the origin O_i and the axes x_i and y_i



Denavit-Hartenberg Convention (1)

- ▶ The *common normal* to two lines is the line containing the minimum distance segment between them.
 1. Uniquely defined if the lines are not co-planar
 2. Infinite number of common normals if the lines are co-planar and parallel
 3. One point (degenerate situation) if the lines are planar and intersect
- ▶ The origin O_i is placed at the intersection between the common normal and the axis z_i
- ▶ Axis x_i is chosen along the common normal in the direction from axis z_{i-1} to z_i
- ▶ y_i is chosen so that the frame is right-handed $\mathbf{y}_i = \mathbf{z}_i \times \mathbf{x}_i$,



Denavit-Hartenberg Convention (2)

Indeterminate Conditions

- ▶ For Frame 0 only z_0 is specified. The origin O_0 and the axis x_0 can be arbitrary.
- ▶ For Frame n there is no joint $n + 1$. So z_n is not uniquely defined. Axis x_n remains normal to axis z_{n-1} . Being joint n typically revolute, z_n is aligned with z_{n-1}
- ▶ When two axes are parallel, there is no unique common normal
- ▶ When two axes intersect there is not a unique definition for axis x_i
- ▶ When joint i is prismatic, the direction of z_{i-1} is arbitrary

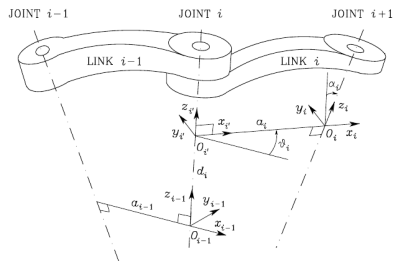
In all these cases, we make a choice to simplify the procedure (i.e. choosing parallel axes for consecutive frames).

Denavit-Hartenberg Convention (3)

Let $O_{i'} - x_{i'} y_{i'} z_{i'}$ be a frame with the axis $z_{i'}$ aligned with z_{i-1} , with axis $x_{i'}$ aligned with x_i and with the origin on the intersection between the common normal and z_{i-1} .

Introduce the following parameters.

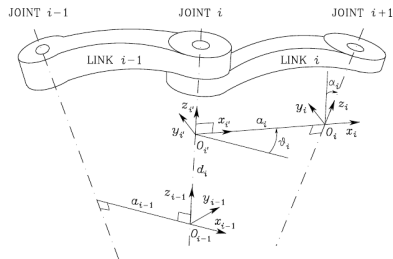
- ▶ a_i distance between O_i and $O_{i'}$
- ▶ d_i coordinate of $O_{i'}$ along z_{i-1}
- ▶ α_i rotation angle between z_{i-1} and z_i about x_i (positive if counter-clock wise)
- ▶ θ_i rotation angle between x_{i-1} and x_i about z_{i-1} (positive if counter-clock wise)



Denavit-Hartenberg Convention (4)

Fixed parameters a_i and α_i . Related to the geometry of the links.

Variable parameters: θ_i for revolute joints, d_i for prismatic joints.

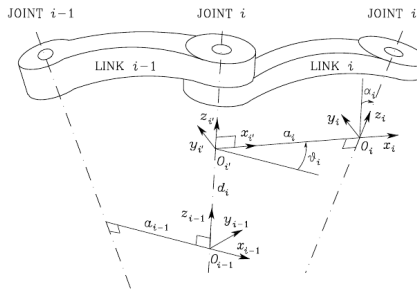


Denavit-Hartenberg Convention (5)

Construction of the transformation matrix between link $i - 1$ and link i .

- ▶ Start from a frame aligned with Frame $i - 1$
- ▶ Translate the frame by d_i along axis z_{i-1} and rotate it by θ_i about axis z_{i-1} align it with frame i' :

$$A_{i'}^{i-1} = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} & 0 & 0 \\ s_{\theta_i} & c_{\theta_i} & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



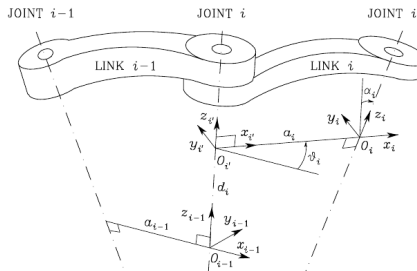
Denavit-Hartenberg Convention (6)

- Translate the frame aligned with Frame i' by a_i along axis $x_{i'}$ and rotate it by α_i about axis $x_{i'}$ aligning it with Frame i :

$$A_i^{i'} = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- The complete transformation is then given by:

$$A_i^{i-1}(q_i) = A_{i'}^{i-1} A_i^{i'} = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} c_{\alpha_i} & s_{\theta_i} s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i} c_{\alpha_i} & -c_{\theta_i} s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Summary of the procedure (1)

1. Find the joint axes and assign $z_0, \dots, z_{n-1}$
2. Complete Frame 0 by choosing x_0, y_0 so that the frame is right-handed (useful to have it coincident with the frame's basis)
3. For all $i \in \{1, \dots, n-1\}$
 - 3.1 locate the origin O_i at the intersection between z_i and z_{i-1} . If axes z_{i-1} and z_i are parallel then: 1. if joint i is revolute locate O_i so that $d_i = 0$, 2. if joint i is prismatic locate O_i at a meaningful position for the range (e.g., the mechanical limit).
 - 3.2 Choose axis x_i along the common normal between axes z_{i-1} and z_i , pointing from Joint i to Joint $i+1$
 - 3.3 Choose axis y_i so that the frame is right handed ($y_i = z_i \times x_i$).

Summary of the procedure (2)

4. Choose Frame n . If Joint n is revolute then align z_n with z_{n-1} , if it is prismatic z_n is arbitrary. For the other axes proceed as for the other joints.
5. For all $i \in \{1, \dots, n\}$ for a table with $a_i, d_i, \alpha_i, \theta_i$
6. Compute the homogeneous transformation $A_i^{i-1}(q_i)$ and then form the total transformation

$$T_n^0(\mathbf{q}) = A_1^0 \dots A_n^{n-1}.$$

7. Given T_0^b and T_e^n complete the direct kinematic function:

$$T_e^b(\mathbf{q}) = T_0^b T_n^0 T_e^n.$$

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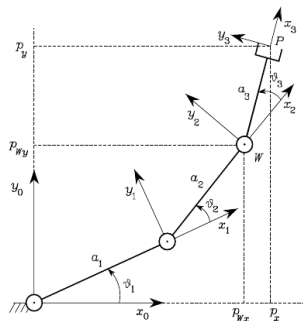
Check quiz

Three Link Planar Arm

- ▶ All joints are revolute and have parallel axes.
 - ▶ The direction of x_0 is arbitrary. For x_1 , x_2 , and x_3 the most convenient choice is to have them aligned with the links.
 - ▶ The d_i parameters are null and the joint variables are essentially the angles between the links.
 - ▶ The D-H table is the following.
- | Link | a_i | α_i | d_i | θ_i |
|------|-------|------------|-------|------------|
| 1 | a_1 | 0 | 0 | θ_1 |
| 2 | a_2 | 0 | 0 | θ_2 |
| 3 | a_3 | 0 | 0 | θ_3 |
- ▶ For all joints we have a rotation about the z_{i-1} axis and a translation of the link length along the direction of the link. Hence

$$A_i^{i-1}(\theta_i) = \begin{bmatrix} c_i & -s_i & 0 & a_i c_i \\ s_i & c_i & 0 & a_i s_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

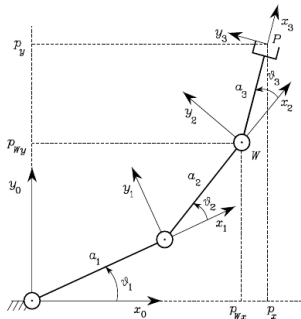
$$T_3^0(\mathbf{q}) = A_1^0 A_2^1 A_3^2 = \begin{bmatrix} c_{123} & -s_{123} & 0 & a_1 c_1 + a_2 c_{12} + a_3 c_{123} \\ s_{123} & c_{123} & 0 & a_1 s_1 + a_2 s_{12} + a_3 s_{123} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Three Link Planar Arm - 1

- Normally, the frame attached to the end effector has z_e as approach angle. So we still need to rotate Frame 3 by 90 degrees about the y_3 axis:

$$T_e^3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



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Spherical Arm

- ▶ The origin O_0 is at the intersection of axis z_0 with axis z_1 (hence $d_1 = 0$). Likewise, The origin of Frame 2 is at the intersection between z_1 and z_2 .

- ▶ The table of the DH parameters is:

Link	a_i	α_i	d_i	θ_i
1	0	$-\frac{\pi}{2}$	0	θ_1
2	0	$\frac{\pi}{2}$	d_2	θ_2
3	0	0	d_3	0

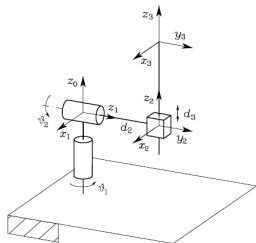
- ▶ The transformation matrices are the following:

Rotate θ_1 about z_0 and $-\frac{\pi}{2}$ about x'_0

$$A_1^0(\theta_1) = \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Translate d_2 along z_1 , rotate θ_2 about z_1 , and then $\frac{\pi}{2}$ about x'_1 :

$$A_2^1(\theta_2) = \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ s_2 & 0 & -c_2 & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Spherical Arm - 1

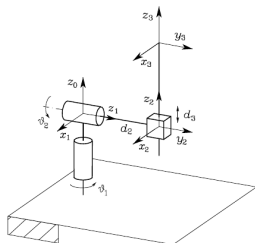
- The Final transformation is a mere translation along z_2

$$A_3^2(d_3) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

which leads to

$$T_3^0(\mathbf{q}) = \begin{bmatrix} c_1 c_2 & -s_1 & c_1 s_2 & c_1 s_2 d_3 - s_1 d_2 \\ s_1 c_2 & c_1 & s_1 s_2 & s_1 s_2 d_3 + c_1 d_2 \\ -s_2 & 0 & c_2 & c_2 d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ d_3 \end{bmatrix}$$



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Anthropomorphic Arm

The arm is a two-link plan arm with an additional rotation about an x-axis.

- ▶ Axes z_0 and z_1 intersect, while z_1 and z_2 are parallel.

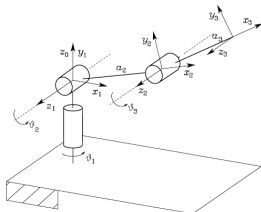
- ▶ The DH parameters are as in the following:

Link	a_i	α_i	d_i	θ_i
1	0	$\frac{\pi}{2}$	0	θ_1
2	a_2	0	0	θ_2
3	a_3	0	0	θ_3

- ▶ The homogeneous transformation are actually a combination of the one seen for the spherical arm and the one for the planar arm.

$$A_1^0(\theta_1) = \begin{bmatrix} c_1 & 0 & s_1 & 0 \\ s_1 & 0 & -c_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_i^{i-1}(\theta_i) = \begin{bmatrix} c_i & -s_i & 0 & a_i c_i \\ s_i & c_i & 0 & a_i s_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \text{ for } i = 2, 3$$

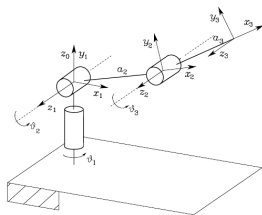


Anthropomorphic Arm - 1

The total transformation is given by

$$T_3^0(\mathbf{q}) = A_1^0 A_2^1 A_3^2 = \begin{bmatrix} c_1 c_{23} & -c_1 s_{23} & s_1 & c_1 (a_2 c_2 + a_3 c_{23}) \\ s_1 c_{23} & -s_1 s_{23} & -c_1 & s_1 (a_2 c_2 + a_3 c_{23}) \\ s_{23} & c_{23} & 0 & a_2 s_2 + a_3 s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$



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Spherical Wrist

- ▶ For this we start the numbering from 3 because it is a wrist (mounted on the tip of a manipulator)
- ▶ Once z_3, z_4, z_5 and x_4 and x_5 are determined arbitrarily.
- ▶ The DH parameters are in the following.

Link	a_i	α_i	d_i	θ_i
4	0	$-\frac{\pi}{2}$	0	θ_4
5	0	$\frac{\pi}{2}$	0	θ_5
6	0	0	d_6	θ_6

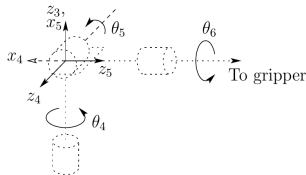
- ▶ The transformation matrices are the following:

Rotate θ_4 about z_3 and $-\frac{\pi}{2}$ about x'_3 :

$$A_4^3(\theta_4) = \begin{bmatrix} c_4 & 0 & -s_4 & 0 \\ s_4 & 0 & c_4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotate θ_5 about z_4 and $-\frac{\pi}{2}$ about x'_4 :

$$A_5^4(\theta_5) = \begin{bmatrix} c_5 & 0 & -s_5 & 0 \\ s_5 & 0 & c_5 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Spherical Wrist - 1

- The transformation matrices for the last link:

Translate of d_6 about z_5

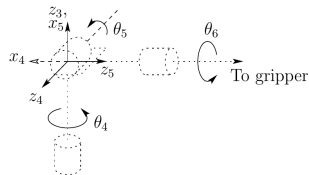
Rotate θ_6 about z_5 and $-\frac{\pi}{2}$ about x_5' :

$$A_5^6(\theta_6) = \begin{bmatrix} c_6 & 0 & -s_6 & 0 \\ s_6 & 0 & c_6 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Final Result :

$$T_6^4(\mathbf{q}) = \begin{bmatrix} c_4 c_5 c_6 - s_4 s_6 & -c_4 c_5 s_6 - s_4 c_6 & c_4 s_5 & c_4 s_5 d_6 \\ s_4 c_5 c_6 + c_4 s_6 & -s_4 c_5 s_6 + c_4 c_6 & s_4 s_5 & s_4 s_5 d_6 \\ -s_5 c_6 & s_5 s_6 & c_5 & c_5 d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q} = \begin{bmatrix} \theta_4 \\ \theta_5 \\ \theta_6 \end{bmatrix}$$



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Operational space

- ▶ The operations of a manipulator are typically specified in terms of position and orientation of the end-effector
- ▶ This requires three coordinates for the position $\mathbf{p}_e$ and the three vectors defining the frame attached to the end effector ($\mathbf{n}_e, \mathbf{s}_e, \mathbf{a}_e$)
- ▶ However, providing nine elements is redundant because they are bound to be orthonormal.
- ▶ It is much better to adopt a non-redundant notation, such as Euler's angles:

$$\mathbf{x}_e = \begin{bmatrix} \mathbf{p}_e \\ \boldsymbol{\Phi}_e \end{bmatrix}$$

- ▶ The vector $\mathbf{x}_e$ is specified in the space where the end-effector operates (operational space).

Joint space

- The joint space (or configuration space) denote the space in which the vector

$$\mathbf{q} = \begin{bmatrix} q_1 \\ \dots \\ q_n \end{bmatrix}$$

is defined, where

$$q_i = \begin{cases} \theta_i & \text{revolute joint} \\ d_i & \text{prismatic joints.} \end{cases}$$

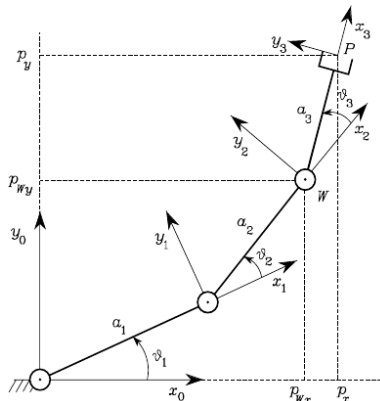
- The direct kinematics equations are defined by a function

$$\mathbf{x}_e = k(\mathbf{q}).$$

Example

It is easy to see that the direct equation are in this case defined by:

$$\mathbf{x}_e = \begin{bmatrix} p_x \\ p_y \\ \phi \end{bmatrix} = \mathbf{K}(q) = \begin{bmatrix} a_1 c_1 + a_2 c_{12} + a_3 c_{123} \\ a_1 s_1 + a_2 s_{12} + a_3 s_{123} \\ \phi_1 + \phi_2 + \phi_3 \end{bmatrix}$$



Workspace

- ▶ The workspace is a key performance metric for the robot
- ▶ It is the region described by the origin of the frame attached to the end-effector for all possible motions at the joints
- ▶ We distinguish between
 - ▶ *Reachable space*: all points the end-effector can reach with at least one configuration
 - ▶ *Dexterous space*: all points the end effector can reach with more than one configuration
 - ▶ Clearly the Dexterous Space is a subspace of the reachable space

Reachable Workspace

- ▶ The reachable workspace can be computed by simply considering the evolution of

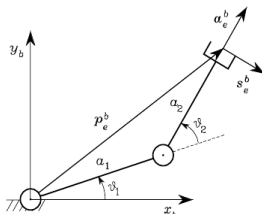
$$\mathbf{p}_e = \mathbf{p}_e(\mathbf{q}), \quad q_{im} \leq q_i \leq q_{iM}, \forall i \in \{1, \dots, n\}.$$

where each joint variable is made to vary within its minimum q_{im} and maximum q_{iM} boundaries.

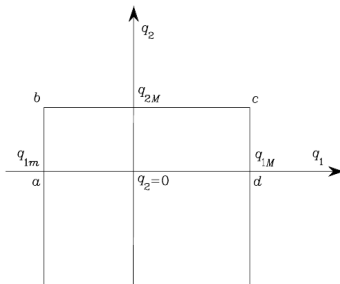
- ▶ The volume described in this way is finite, closed, and connected (for the continuity of $\mathbf{p}_e(\mathbf{q})$); thus its frontier (the bordering surface) becomes very important.

Example

Consider the two link planar manipulator

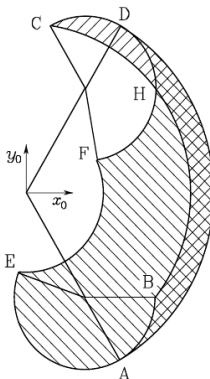
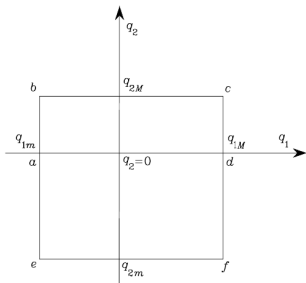


The two link variables take values in a rectangular region



Example (continued)

The best way to construct the workspace is to make the arm move following the perimeter of the region of ammissible configurations: Segments ab , bc , cd , da , ae , ef , fd . We show the trajectory followed by the end-effector.



Outline

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Direct Kinematics

The Denavit-Hartenberg Convention

Examples

- Three-link Planar Arm

- Spherical Arm

- Anthropomorphic Arm

- Spherical Wrist

Joint Space and Operational Space and Work Space

Useful definitions

Check quiz

Accuracy

- ▶ In the expression of the direct kinematics

$$\mathbf{x}_e = k(\mathbf{q}),$$

we have not explicitly shown the role of the DH parameters (e.g., the lengths of the links)

- ▶ These mechanical parameters are known with some degree of inaccuracy (due to mechanical tolerance).
- ▶ Therefore when we set a value for the joint parameters $\mathbf{q}$, we will have some possible deviation in $\Delta\mathbf{x}_e$ in the position of the end-effector
- ▶ This deviation is called accuracy and is usually less than 1mm.
- ▶ The accuracy depends on the geometry of the manipulator (what is the maximum reach) and on the position in the workspace

Repeatability

- ▶ Repeatability is a measure of the manipulator's ability to return to a previously visited position.
- ▶ This parameter is very important for teach-by-showing examples and it depends also on the precision of transducers and controllers
- ▶ For a manipulator with a maximum reach of 1.5m, accuracy is usually in the range $[0.2, 1]$ mm while repeatability is in the range $[0.02, 0.2]$ mm.

Kinematic Redundancy

- ▶ A manipulator is has kinematic redundancy if its number of DoFs is greater than the number of variables needed to describe a task.
- ▶ An intrinsic redundancy exists when the dimension n of the joint space is larger than the dimension m of the operational space.
- ▶ In general if the task requires a specification of r dimension ($r \leq m$) we have redundancy of $r < n$.
- ▶ As an example, the 3 DoF planar manipulator ($n = 3$) is redundant if we only impose the position of the end effector $r = 2$, it is not if we impose also the orientation $r = 3$
- ▶ Redundancy is related to the task, and not to the system itself.
- ▶ Redundant manipulator are often used to increase the dexterity or the mobility (e.g., the human arm has 7 DoF)

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Question 1

- ▶ What is the main purpose of *direct kinematics*?
 1. To find the pose of the end-effector given the joint variables.
 2. To find the joint variables given the pose of the end-effector.
 3. To determine the manipulator's workspace.
 4. To calculate the power consumption of the robot.

Answer 1

The correct answer is **1. To find the pose of the end-effector given the joint variables.**

- ▶ Direct kinematics is the forward mapping from the joint space to the operational space.
- ▶ Answer 2 describes *inverse kinematics*.

Question 2

- ▶ What is the primary function of the **Denavit-Hartenberg (D-H) Convention**?
 1. To calculate the torque at each joint.
 2. To provide a systematic way to assign coordinate frames to each link.
 3. To determine the speed of the end-effector.
 4. To simplify the construction of the robot arm itself.

Answer 2

The correct answer is **2. To provide a systematic way to assign coordinate frames to each link.**

- ▶ The D-H convention is a standard for establishing a coordinate frame on each link of an open kinematic chain, which is essential for deriving the kinematic equations.

Question 3

- A manipulator is considered *kinematically redundant* when:
1. Its repeatability is higher than its accuracy.
 2. The number of its degrees of freedom is greater than the number of variables needed to describe a task.
 3. It has both revolute and prismatic joints.
 4. It can reach every point in its workspace.

Answer 3

The correct answer is **2**. **The number of its degrees of freedom is greater than the number of variables needed to describe a task.**

- ▶ Redundancy allows for greater dexterity and the ability to avoid obstacles, as there are multiple joint configurations to achieve the same end-effector pose.

Question 4: D-H Parameters

- Given two consecutive revolute joints with *parallel* axes, what is the value of the Denavit-Hartenberg parameter α_i ?
1. 90°
 2. 0°
 3. 45°
 4. 180°

Answer 4

The correct answer is **2.** 0° .

- ▶ The parameter α_i represents the *twist angle*, which is the angle between the z_{i-1} and z_i axes.
- ▶ For parallel axes, the angle between them is always 0° .
- ▶ This specific condition leads to an indeterminate choice for the x_i axis, but the value of α_i itself is clearly defined.

Question 5: D-H Transformation Order

- The homogeneous transformation matrix A_i^{i-1} from link $i - 1$ to link i is a product of four simpler transformations. What is the *correct order* of these transformations according to the D-H convention?
1. Translation along x_i , rotation about x_i , translation along z_{i-1} , rotation about z_{i-1}
 2. Rotation about z_{i-1} , translation along z_{i-1} , translation along x_i , rotation about x_i
 3. Rotation about x_i , translation along x_i , rotation about z_{i-1} , translation along z_{i-1}
 4. Translation along z_{i-1} , rotation about z_{i-1} , rotation about x_i , translation along x_i

Answer 5

The correct answer is **2. Rotation about z_{i-1} , translation along z_{i-1} , translation along x_i , rotation about x_i .**

- The sequence of operations is:
1. Rotate about z_{i-1} by angle θ_i .
 2. Translate along z_{i-1} by distance d_i .
 3. Translate along x_i by distance a_i .
 4. Rotate about x_i by angle α_i .

Question 6: Workspace Concepts

- ▶ What is the primary reason the *dexterous workspace* is a more practical measure of a manipulator's capability than its reachable workspace?
 1. It includes all possible orientations of the end-effector.
 2. It is easier to calculate and visualize.
 3. It accounts for the forces the manipulator can exert.
 4. It represents the region where the end-effector can be placed with more flexibility.

Answer 6

The correct answer is **4. It represents the region where the end-effector can be placed with more flexibility.**

- ▶ The dexterous workspace is the subset of the reachable workspace where a point can be reached using *multiple joint configurations*.
- ▶ This redundancy in configurations allows the robot to perform tasks with greater agility and avoid obstacles.

Question 7: Accuracy vs. Repeatability

- A manipulator with high *repeatability* but low *accuracy* would exhibit which of the following behaviours?
1. It consistently returns to the same position, but that position is far from the commanded target.
 2. It reaches the commanded target position every time, but with a wide scattering of final locations.
 3. It can only move very slowly.
 4. It fails to reach the target and is also inconsistent in its final position.

Answer 8

The correct answer is **1. It consistently returns to the same position, but that position is far from the commanded target.**

- ▶ *Repeatability* is a measure of the consistency of the robot's motion.
- ▶ *Accuracy* is a measure of how close the robot's end-effector is to a commanded position.

Question 8: Non-redundant Manipulator

- ▶ For a 6 DoF non-redundant serial manipulator, what is the dimension of the operational space for a task that requires both position and orientation control?
 1. 3
 2. 4
 3. 6
 4. It is dependent on the type of joints.

Answer 8

The correct answer is **3. 6.**

- ▶ To fully specify the pose (position and orientation) of an end-effector in 3D space, you need 3 variables for position and 3 for orientation, for a total of 6 degrees of freedom.